

Translated

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Pick's Theorem

A polygon without self-intersections is called lattice if all its vertices have integer coordinates in some 2D grid. Pick's theorem provides a way to compute the area of this polygon through the number of vertices that are lying on the boundary and the number of vertices that lie strictly inside the polygon.

Formula

Given a certain lattice polygon with non-zero area.

We denote its area by S , the number of points with integer coordinates lying strictly inside the polygon by I and the number of points lying on polygon sides by B .

Then, the **Pick's formula** states:

$$S = I + \frac{B}{2} - 1$$

In particular, if the values of I and B for a polygon are given, the area can be calculated in $O(1)$ without even knowing the vertices.

This formula was discovered and proven by Austrian mathematician Georg Alexander Pick in 1899.

Proof

The proof is carried out in many stages: from simple polygons to arbitrary ones:

- A single square: $S=1$, $I=0$, $B=4$, which satisfies the formula.
- An arbitrary non-degenerate rectangle with sides parallel to coordinate axes: Assume a and b be the length of the sides of rectangle. Then, $S=ab$, $I=(a-1)(b-1)$, $B=2(a+b)$. On substituting, we see that formula is true.
- A right angle with legs parallel to the axes: To prove this, note that any such triangle can be obtained by cutting off a rectangle by a diagonal. Denoting the number of integral points